FRAME 1 OF 2

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5957

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Hydrogen
See figure 1
Mounting position: Any
Weight: 3-1/2 ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ecc	Ip	egx	egy
Unit:	V	kv	kv	Vdc	Vdc	Aac	v	v
Maximum:	6.9	8.0	8.0	---	---	2.9	200	---
Minimum:	5.7	--- (see note 1)	5% epy (see note 2)	2,500	---	---	---	See note 3
TEST CONDITIONS:	6.3	8.0	---	---	0	---	---	130 (see note 4)

ABSOLUTE RATINGS:

Parameter:	ib	Ib	tk	dik/dt	Pb	TA	prr
Unit:	a	mAdc	sec	a/μs	---	°C	---
Maximum:	83	100	---	1,200	3.0×10^9	90	---
Minimum:	---	---	180	---	---	-50	---
TEST CONDITIONS:	---	---	180	---	---	---	4,500

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
3241	Heater current		0.65	II	If	5.5	6.7	Aac
3267	Instantaneous starting	epy = 8.0 kv (min); Ef = 6.9 Vac (see notes 6 and 8)	0.65	II	---	---	---	---
3246	Operation (1)	epy = 9.0 kv; Ef = 5.7 Vac; t = 300 (see note 6)	0.65	II	egy	---	130	v
(D) 3247	DC anode voltage for conduction	Ef = 5.7 Vac	0.65	II	Ebb	---	1,500	Vdc
3246	Operation (2)	epy = 10.0 kv; t = 300; Ef = 5.7 Vac; prr = 3,800 (see note 9)	0.65	II	egy	---	130	v
(D) 3251	Pulse emission (method A)	ik = 90 a (min); prr = 60 ±10%; tp = 5.0 µs ±10%; tr = 0.5 µs (max); starting time of current pulse = 1.0 µs; specified time interval = 4.0 µs	0.65	II	egk	---	150	v
	Quality conformance inspection, part 2							
3246	Operation (1A)	Operation (1), except Ef = 6.9 Vac	---	---	egy	---	130	v
3256	Anode delay time	Operation (1); t = 120	---	---	tad	---	0.5	µs
3256	Anode delay time drift	Anode delay time (see note 5)	---	---	Δtad	---	0.1	µs
3261	Time jitter	epy = 3.0 kv (max) (see note 6)	---	---	tj	---	0.01	µs
3246	Operation (2A)	Operation (2), except Ef = 6.9 Vac	---	---	egy	---	130	v
	Quality conformance inspection, part 3							
---	Life test	Group B (see note 7); t = 96 hours "on" and 1 hour "off" (tube mounted in horizontal position); t = 500 hours	---	---	---	---	---	---
---	Life-test end points:							
3246	Operation (1) and (1A)		---	---	egy	---	140	v
3261	Time jitter	Except epy = 140 v	---	---	tj	---	0.02	µs
3247	DC anode voltage for conduction		---	---	Ebb	---	1,000	Vdc
3256	Anode delay time	Except epy = 140 v	---	---	tad	---	0.6	µs
3256	Anode delay time drift	Except epy = 140 v	---	---	Δtad	---	0.1	µs

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(D)	Periodic-check tests							
3246	Operation at elevated ambient temperature	t = 5 hours; TA = 90°C; Ef = 6.9 Vac (see note 6)	---	---	egy	---	130	v
1032	Vibration, mechanical	No voltages Axes of application (see figure 1) Frequency range (10 to 50 Hz); Vibration level (0.080 ±0.005 inch double amplitude) (see note 10)	---	---	---	---	---	---
1042	Shock, specified pulse	Condition A	---	---	---	---	---	---
---	Shock, specified pulse end points:							
3246	Operation (1)		---	---	egy	---	130	v
3247	DC anode voltage for conduction		---	---	Ebb	---	1,500	Vdc
3261	Time jitter		---	---	tj	---	0.01	µs

NOTES:

1. For instantaneous starting applications where anode voltage is applied instantaneously, the maximum permissible epy is 7 kv.
2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 2.5 kv during the first 25 µs after the anode pulse.
3. Driver pulse, measured at tube socket with thyratron grid disconnected; egy = 175 v minimum; time of rise = 0.5 µs maximum; grid pulse duration = 2 µs minimum. Impedance of drive circuit = 1,500-ohms maximum.
4. Drive pulse measured at tube socket with thyratron grid disconnected.
5. During the interval between 2 minutes and 7 minutes of the anode delay time test, the change in anode delay time drift (Δtad) relative to the tad value observed on the anode delay time test shall not exceed the specified value.
6. The anode circuit constants shall be chosen so that epy = 8.0 kv under resonant charging conditions; dik/dt = 1,200 a/µs minimum; ib = 83 a; tp = 0.25 µs; prr = 4,500; inverse voltage due to mismatch = 400 volts maximum. The grid pulse characteristics shall be tp = 2.0 µs maximum; tr = 0.5 µs minimum; driver impedance = 1,500-ohms minimum.
7. The anode circuit constants shall be chosen so that epy = 8.0 kv under resonant charging conditions; ib = 83 a minimum; dik/dt = 1,200 a/µs minimum; tp = 0.25 µs; prr = 4,500; inverse voltage due to mismatch = 1.9 kv minimum. The grid pulse characteristics shall be tp = 2.0 µs maximum; tr = 0.5 µs minimum; driver impedance = 1,500-ohms minimum.
8. Any tube failing to start within three attempts will be considered a failure.
9. The anode circuit constants shall be chosen so that epy = 8.0 kv under resonant charging conditions: dik/dt = 1,000 a/µs minimum; ib = 90 a; tp = 0.50 µs; prr = 3,000. The grid pulse characteristics shall be tp = 2.0 µs maximum; tr = 0.50 µs minimum; driver impedance = 1,500-ohms minimum.
10. There shall be no pronounced resonance over the range from 10 to 50 Hz.

FRAME 7 OF 7

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MI
Air Force - 11, 80
DSA - ES

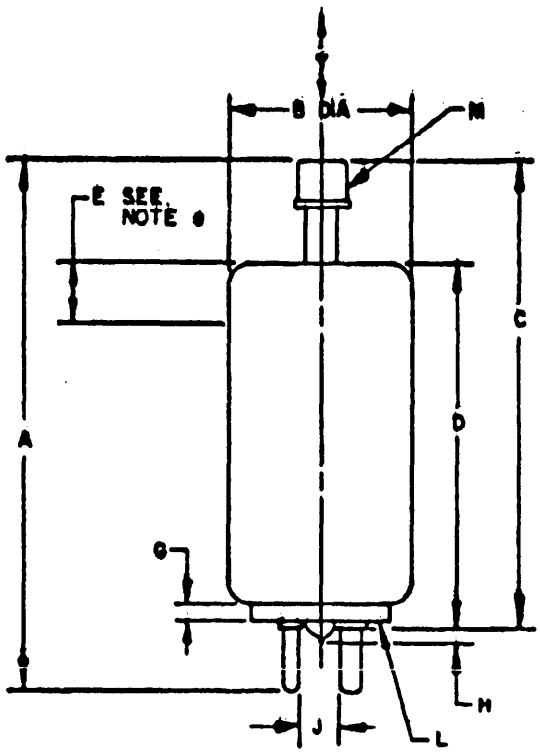
User activities:
Army - AV, WC
Navy - AS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

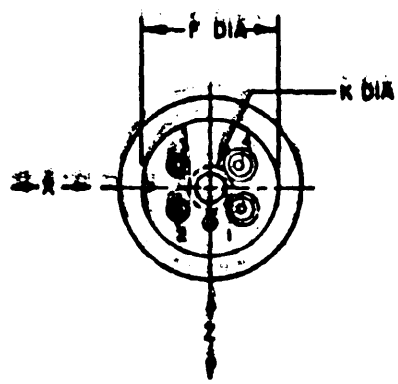
Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		4.455 (113.16)
B	1.440 (36.58)	1.560 (39.62)
C	3.500 (88.90)	3.875 (98.43)
D	2.750 (69.85)	3.000 (76.20)
J		.260 (6.60)
Quality conformance inspection, part 3 (periodic check)		
F		1.312 (33.32)
G	.146 (3.71)	.166 (4.22)
H		.250 (6.35)
L	Base: see note b	
M	Cap: C1-1(EIA)	
Reference dimensions		
E	.750 (19.05)	
K	.312 (7.92)	



PIN CONNECTIONS	
Pin No.	Element
1	h
2	k
3	g
4	h,k
cap	a

NOTES:
a. Clamping area insulated from ground.
b. Pins and pin arrangement as per A4-89(EIA).

FIGURE 1. Outline drawing of electron tube type 5957.

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